FRANCES (SU) LIU

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RESEARCH INTERESTS

My research asks when and how different information sets sharpen volatility forecasts, combining high-frequency market microstructure data and media sentiment within a common HAR-based forecasting framework. Building on this and on prior industry experience in portfolio construction, I am also interested in extending these methods to high-dimensional portfolio applications.

Core areas: financial econometrics; volatility modeling and forecasting; high-frequency and tick-level data; extended trading hours and information transmission; machine learning and deep sequence models in asset pricing; media sentiment and state-dependent predictability; systemic risk and dependence modeling.

EDUCATION

Doctor of Philosophy (PhD) in Finance *2024.02–2027.08 (expected)*
University of Technology Sydney (UTS), Sydney, Australia

Supervisors: Assoc. Prof. Vitali Alexeev (UTS, Finance) & Assoc. Prof. Katja Ignatieva (UNSW, Actuarial)

Thesis: *[title not yet finalised]*

Master of Science in Financial Engineering *2015.08–2017.06*
Stevens Institute of Technology, New Jersey, United States *GPA: 3.9/4.0*

Thesis: Systemic Risk Measurement: CoVaR and Copula Models (ARMA/TGARCH conditional moments, copula-based dependence between financial sectors, time-series CoVaR). Supervised by Assoc. Prof. Zhenyu Cui (School of Systems & Enterprises, Hanlon Lab); awarded the Academic Excellence Award

Bachelor of Economics in Financial Engineering *2011.09–2015.06*
Harbin University of Commerce, Harbin, China

WORKING PAPERS

Reading Between the Trades: Neural Encoding Extended-Hours Tick Sequences for Volatility Forecasting
with Vitali Alexeev (UTS), Adam Clements (QUT), and Katja Ignatieva (UNSW) *working paper*

- Replaces the coarse overnight aggregates used in the volatility literature with the complete tick-level record of each extended-hours session, compressing variable-length, irregularly spaced transaction sequences into fixed-dimensional representations through neural encoders and integrating them with the HAR model
- An incremental architecture hierarchy isolates the contribution of each modeling ingredient; applied to all DJIA constituents over 2002–2025, the simplest architecture reduces out-of-sample forecast loss by approximately 11% at the one-day horizon, roughly twice the gain of the best scalar overnight proxy
- Pre-open and post-close sessions deliver comparable aggregate gains through opposite channels: 64% of the pre-open improvement is recoverable from a single within-session aggregate, while 81% of the post-close improvement is attributable to tick-sequence structure. Gains are stable across the Global Financial Crisis and COVID-19 episodes
- *JEL*: C45, C53, G12, G14

When Does Sentiment Predict Volatility?
with Vitali Alexeev (UTS) and Katja Ignatieva (UNSW) *under review, Journal of Applied Econometrics*

- Augmenting HAR models with Thomson Reuters MarketPsych sentiment yields only modest average improvements, but this average masks strong state dependence
- Sentiment becomes materially more informative in the upper tail of the volatility distribution and, more importantly, when volatility persistence weakens; persistence is measured within a fractional integration framework
- Out-of-sample gains concentrate in regimes jointly characterised by high volatility and low persistence, indicating

that sentiment supplies forward-looking narrative information precisely when price-based models are least reliable
- *JEL*: C22, C53, G12, G14

CONFERENCE PRESENTATIONS

When Does Sentiment Predict Volatility? (IAAE Annual Conference, Lisbon) 2026

[add SoFiE, Bachelier World Congress, and GLO Asia only if you presented there; if you attended without presenting, list them under a separate "Conferences Attended" heading or leave them off]

AWARDS AND SCHOLARSHIPS

UTS Business Doctoral Scholarship (full stipend) 2024–2027
UTS International Research Scholarship (full tuition) 2024–2027
Academic Excellence Award, Stevens Institute of Technology 2017

RESEARCH EXPERIENCE

University of Technology Sydney *Sydney, Australia*
Doctoral Researcher, Finance Discipline Group 2024.02–present
- Constructed a full research pipeline over tick-level transaction data for the DJIA universe, 2002–2025: data cleaning and normalization, realized volatility construction, neural sequence encoders in PyTorch, and rolling out-of-sample evaluation with strict look-ahead controls
- Built and ran the estimation pipeline on GPU cluster infrastructure, with rolling re-estimation and a density-aware gate for sparse sessions
- Traced forecast gains to their economic source: order-flow volume dominates in both sessions, while inter-trade timing contributes disproportionately pre-open
- Extended classical econometric frameworks with quantile regression, fractional integration, and machine learning models, and derived corrections improving estimator accuracy in volatility models

TEACHING EXPERIENCE

University of Technology Sydney *Sydney, Australia*
Tutor, Finance Discipline Group 2024–present
- *Financial System* (tutorials)
- *Investment Analysis* (tutorials)

[add semesters taught, cohort level, and teaching evaluation scores if strong]

INDUSTRY EXPERIENCE

Seven years applying, in industry, the same idea-to-deployment process the PhD now uses in research.

Webull Financial LLC *Changsha, China*
Senior Quantitative Analyst 2020.04–2023.11
- Built factor models for US equities, mining stock-level factor data for standalone strategies and for a factor pool feeding machine learning pipelines
- Developed multivariate time-series models across Chinese and US equity index universes, with predictive modeling, statistical analysis, optimization, and data quality calibration
- Designed and co-developed the quantitative research database and strategy backtesting environment used across the quantitative team

Pricing and Analytics (Shenzhen) Limited *Shenzhen, China*
Quantitative Analyst: Derivatives Pricing and Valuation 2017.12–2019.10
- Designed pricing and valuation models for mortgage-backed products including REITs and CLOs; both models

were registered as software copyrights
- Applied Black-Scholes, risk-neutral valuation, and martingale pricing methods, and implemented extreme value theory and Value at Risk for portfolio tail risk

TECHNICAL SKILLS

Programming: Python (PyTorch), R, C++, SQL
Computing: GPU cluster experiment orchestration; large-scale rolling out-of-sample pipelines
Econometrics: volatility modeling (HAR), realized volatility, quantile regression, fractional integration, copulas, extreme value theory
Machine Learning: sequence encoders (attention, temporal convolution, GRU, Deep Sets) for irregular, variable-length data
Data: tick-level transaction data
Languages: English (proficient), Cantonese (proficient), Mandarin (native)

REFEREES

Assoc. Prof. Vitali Alexeev, Finance, University of Technology Sydney (PhD supervisor)
Assoc. Prof. Katja Ignatieva, Actuarial Studies, UNSW (PhD co-supervisor)
Assoc. Prof. Zhenyu Cui, Financial Engineering, Stevens Institute of Technology (master's project supervisor)

[Confirm each referee is willing before listing, and add their email addresses]